

AI AGRONOMY REPORT: SECTOR 2 (SOUTH RIDGE)

Generated by Vita-Core Sentinel AI Engine (RAG-Local Fallback Model)

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📄 Executive Summary

A comprehensive soil analysis has been compiled for **Sector 2 (South Ridge)** containing **8.2 acres** of **WHEAT**.

By correlating the visual bioluminescent glow metrics from the sensor ribbon with our local reference catalog, we have generated current soil indices.

* **Overall Soil Health:** 🚨 CRITICAL STRESS

* **Active Soil Parameters:**

* **Nitrogen (N):** 175.1 mg/kg

* **Soil Moisture:** 65%

* **Acidity/Alkalinity (pH):** 5.2

📈 Historical 7-Day Trend Analysis

Our database has analyzed the latest visual drone images over the past week:

- 2026-06-27: N=175.1mg/kg, Moisture=65%, pH=5.2
- 2026-06-27: N=210.8mg/kg, Moisture=22.4%, pH=7
- 2026-06-23: N=89.6mg/kg, Moisture=56.6%, pH=7.1
- 2026-06-22: N=92.6mg/kg, Moisture=51.3%, pH=7.2
- 2026-06-21: N=171.1mg/kg, Moisture=46.6%, pH=6.9
- 2026-06-20: N=149.7mg/kg, Moisture=49.4%, pH=6.6
- 2026-06-19: N=146.1mg/kg, Moisture=54.7%, pH=6.5

* **Assessment:** The moisture levels have showed a critical downward vector over the past 48 hours. Nitrogen levels are depleting due to high vegetative demands, requiring immediate remediation.

💧 Exact Restoration Quantities & Calculations

To restore the optimal soil equilibrium needed for this specific crop, apply the following quantities:

💧 1. Moisture & Irrigation Restoration

* **Current Deficiency:** Target Min Moisture - Current = 65.0% vs optimal range.

* **Exact Water Volume:** 0 Liters**

* **Action Plan:** Trigger irrigation systems for Sector 2 immediately. Distribute the water evenly over the root-zone.

🌱 2. Nitrogen Fertilization

* **Current Deficiency:** Target Min Nitrogen - Current = 175.1 mg/kg.

* **Exact Urea quantity:** 0 kg**

* **Action Plan:** Apply Urea granules (46% N) at a uniform density across the acreage. Do not over-apply to prevent chemical runoffs into surrounding irrigation channels.

⚖️ 3. pH Balancing

Soil Acidity Warning: Current soil pH is 5.2 (Optimal: 6-7.5). Apply 4100 kg of Agricultural Limestone (calcium carbonate) to neutralize soil acidity.

👨‍🌾 Agronomist Field Recommendations

1. **Irrigate Immediately:** Delaying watering by more than 24 hours will trigger crop cell wall damage.
2. **Stagger Nitrogen Applications:** Apply 50% of the calculated Urea immediately, followed by the remaining 50% post-watering.
3. **Scheduled Rescanning:** Schedule a drone sensor flight tomorrow at 9:00 PM to capture the next bioluminescent wavelength return.